

# CS240 Principles of Data Science

หลักการวิทยาการข้อมูล

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## 1. Introduction

# Topics

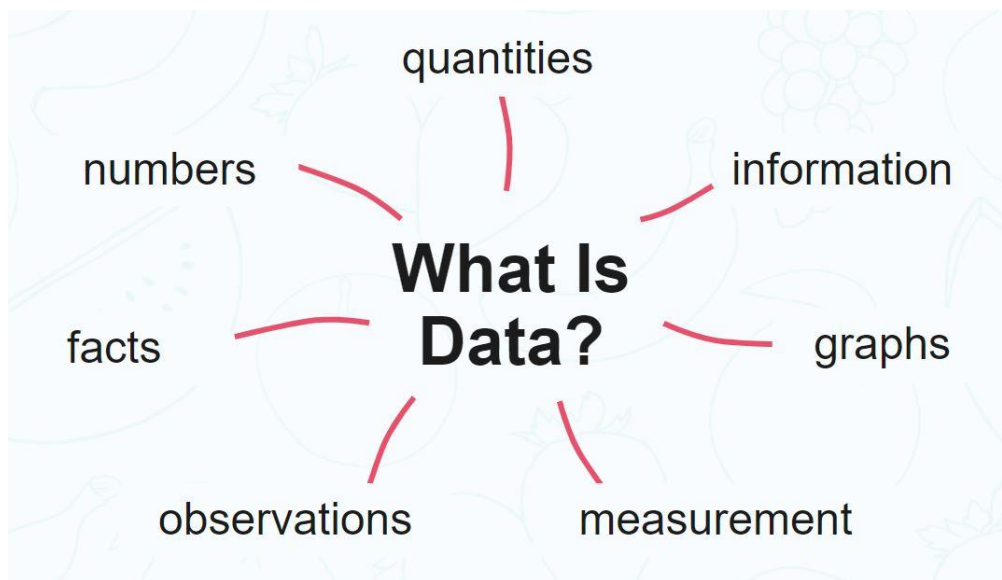
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- What is Data Science?
- Applications
- Data Science Process
- Introduction to Google CoLab

# What is Data?

*You can have data without information, but you cannot have information without data.*

- Data are units of information (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation.”



<https://www.twinkl.com/teaching-wiki/data>

## Data and Information



**Data**

Data can be defined as a representation of facts, concepts, or instructions in a formalized manner which should be suitable for communication, interpretation, or processing by human or electronic machines.

**VS**



**Information**

Information is organized or classified data, which has some meaningful values for the receiver. Information is the processed data on which decisions and actions are based.

**GetupLearn.com**

# Data is all around us!

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- Lots of data is being collected through text, audio recordings, photography and many more.
  - Social Network
  - Financial transactions, bank/credit transactions
  - Online trading and purchasing
  - Web data, e-commerce

# What is Data Science?

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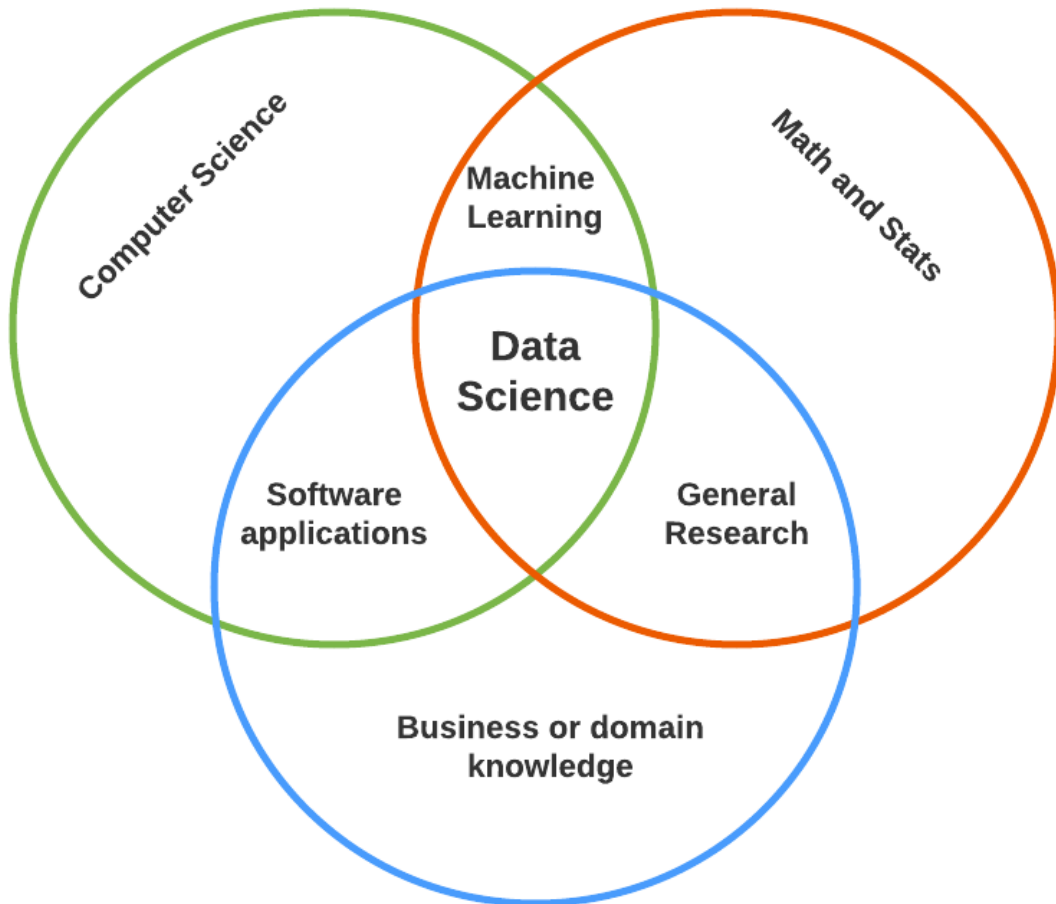
- Data
  - Facts and statistics collected together for reference or analysis.
  - A set of values of subjects with respect to **qualitative or quantitative** variables.
- Science
  - A **systematic enterprise** that builds and organizes knowledge in the form of **testable explanations** and **predictions** about the universe.
- Data Science
  - Uses of scientific methods, processes, algorithms and systems **to extract knowledge and insights from data**.

# Data Science

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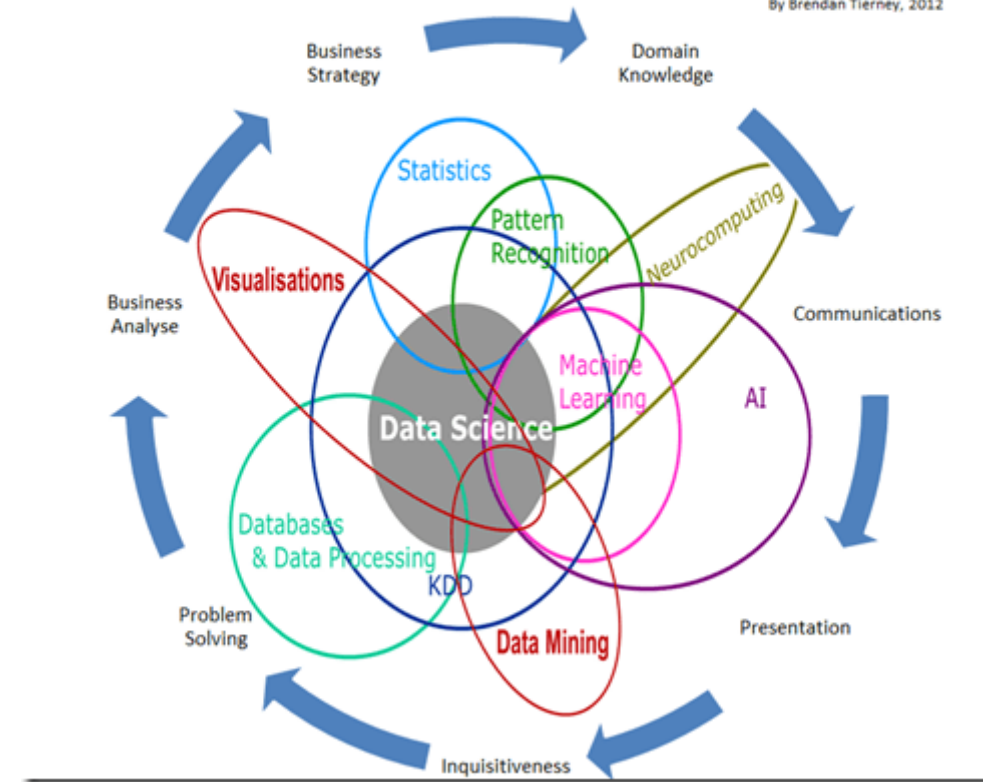
- An area that manages, manipulates, extracts, and interprets knowledge from tremendous amount of data
- Data science (DS) is a multidisciplinary field of study with goal to address the challenges in big data

# Data Science: Related Study Fields



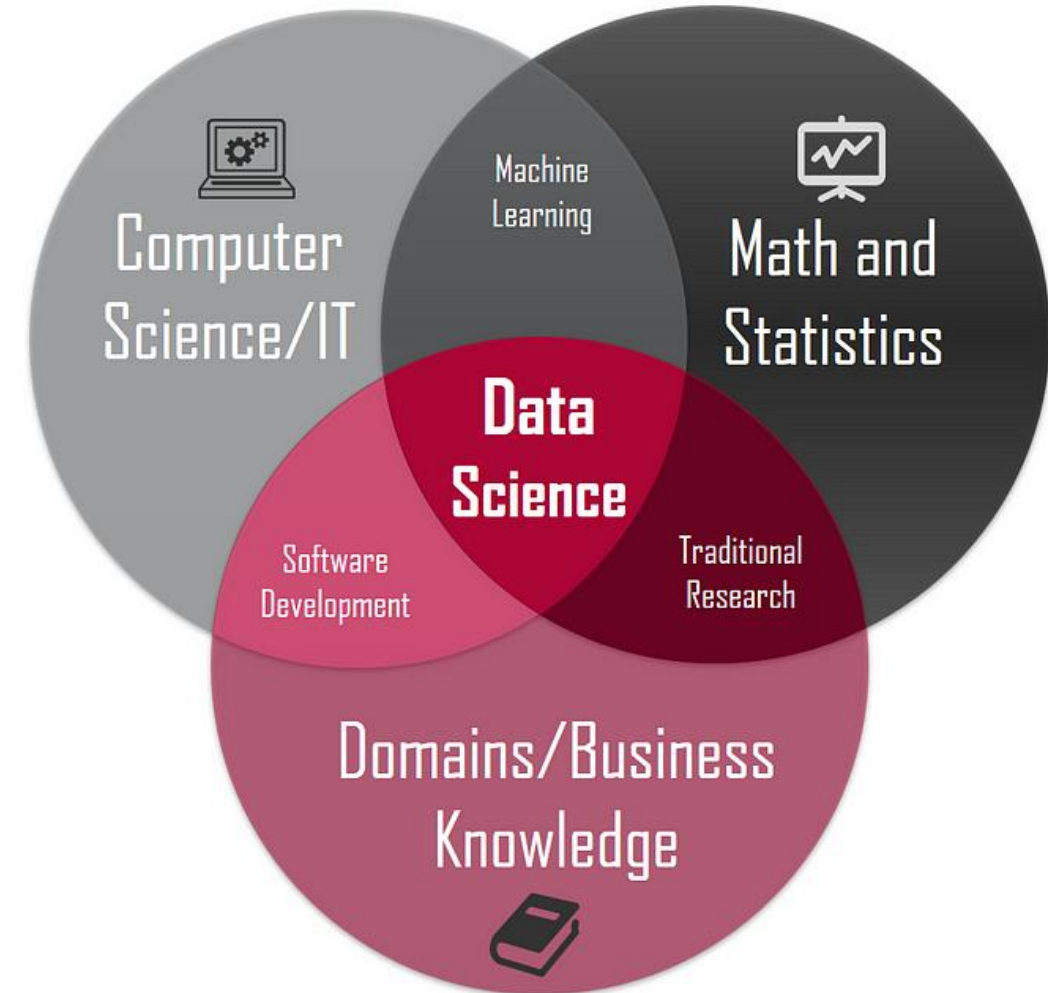
## Data Science Is Multidisciplinary

By Brendan Tierney, 2012



# Data Science: Related Study Fields

- Computer Science
  - Provide tools (Software, Programming languages)
  - Data Engineering
- Mathematics and Statistics
  - Analytics tools (Math and Stat models)
- Domain Expertise
  - Define business problem and objectives
  - Integrate business knowledge into data analysis process
  - Interpret results



# Data Science

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- Theories and techniques from many fields and disciplines are used to investigate and analyze a large amount of data to help decision makers in many industries such as
  - Science
  - Engineering
  - Economics
  - Politics
  - Finance
  - Education

# Applications

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- Computer Science

- Pattern recognition, visualization, data warehousing, High performance computing, Databases, AI

- Mathematics

- Mathematical Modeling

- Statistics

- Statistical and Stochastic modeling, Probability.

# Examples: Applications for Data Science



- Gain customer insights

- You may gather data about a customer each time they visit your website to identify trends in their behaviors.

- Increase security

- Banks use complex ML algorithms to detect fraud based on deviations from a user's typical financial activities.

- Predict future market trends

- Tracking purchase data, celebrities and influencers, and search engine queries can reveal what products people are interested in.

# Data Science VS. Data Analytics

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- Data Science

- Data science is an entire field dedicated to making data more useful.
- A data scientist is a professional that uses raw data to develop new ways to model data and understand the unknown.
- Often, their job responsibilities incorporate various components of computer science, predictive analytics, statistics, and machine learning.
- The collections of information that data scientists work with can be quite large, requiring expertise to organize and navigate.

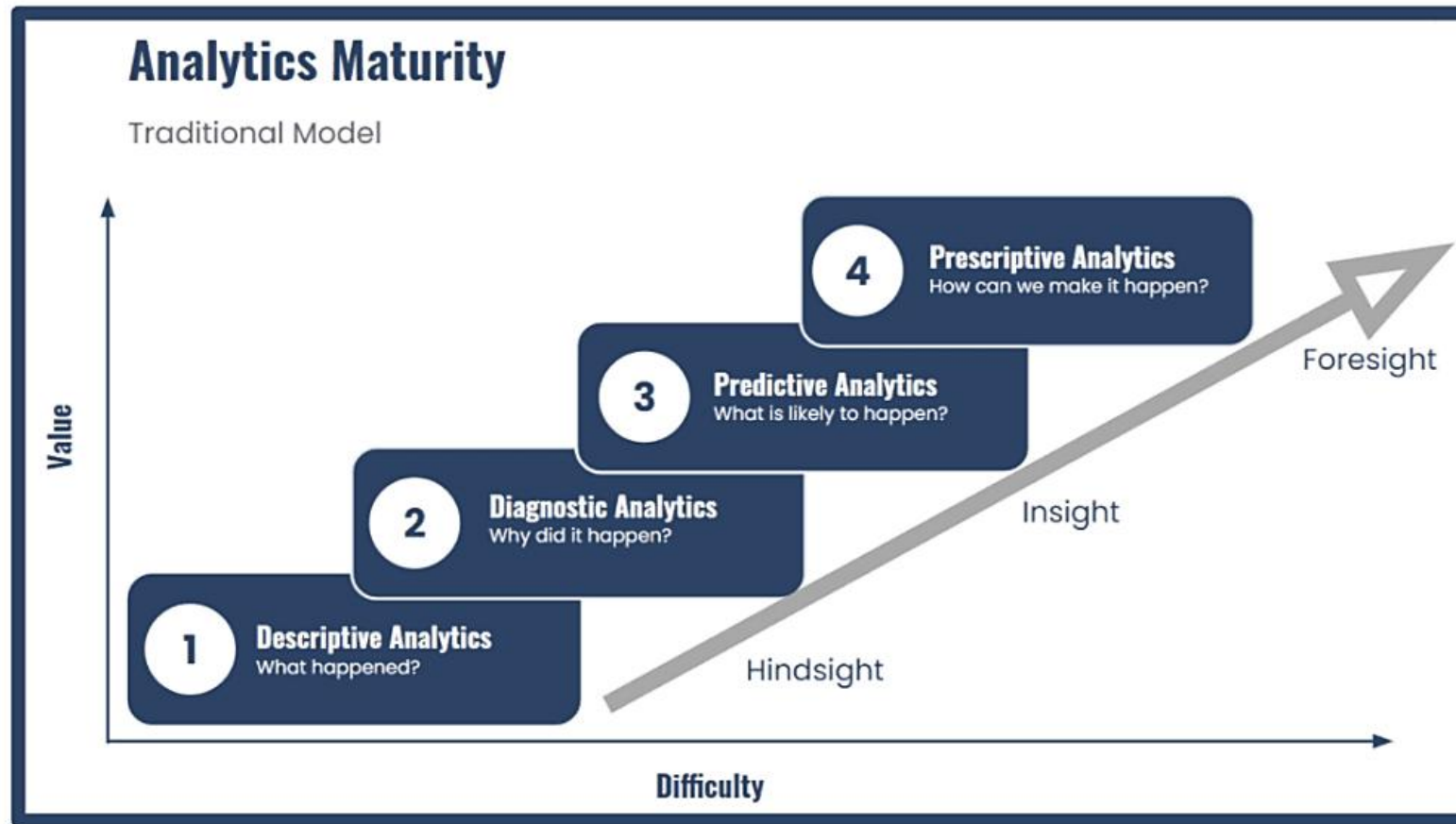
# Data Science VS. Data Analytics

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- Data Analytics

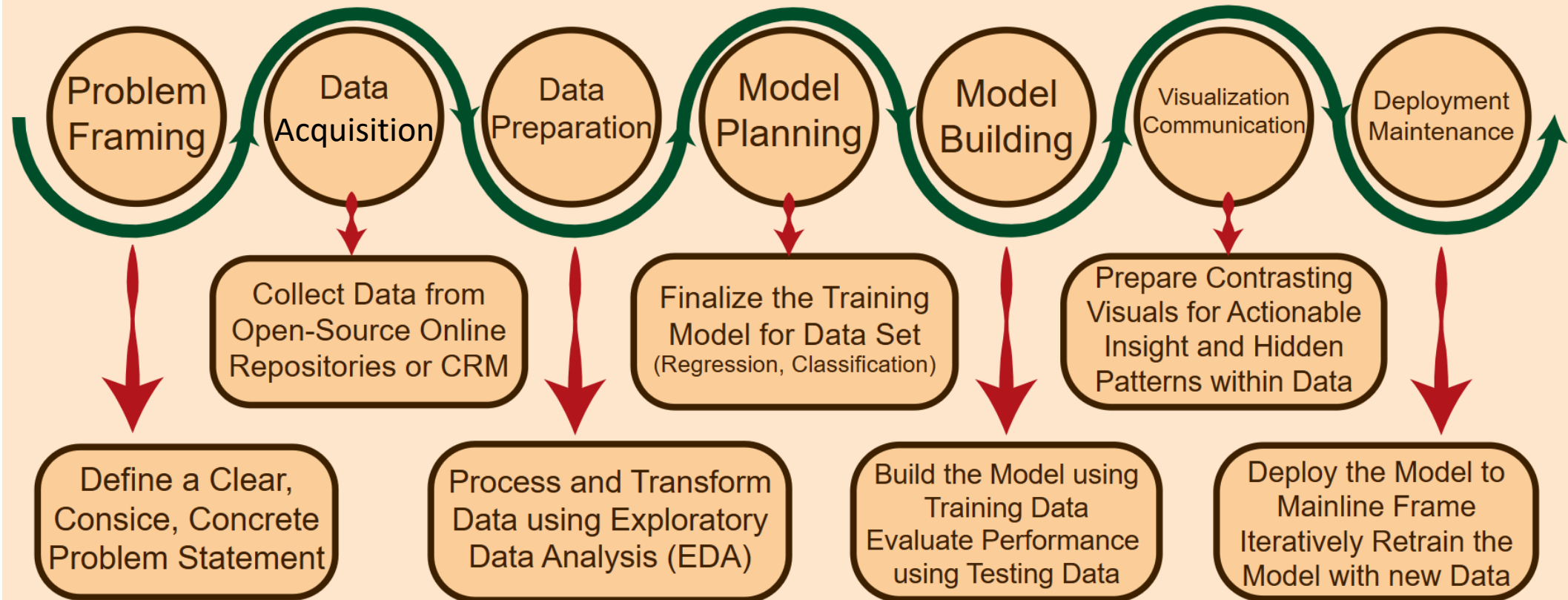
- Data analytics is a subfield of the larger data science discipline.
- The aim of data analytics is to create methods to capture, process, and organize data to uncover actionable insights for current problems.
- Analysts focus on processing the information stored in existing datasets and establishing the best way to present this data.
- Data analysts rely on statistics and data modeling to solve problems and offer recommendations that can lead to immediate improvements.

# Example: Analytics Maturity



# Data Science Process

## DATA SCIENCE PROCESS



# Data Science Process

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- Problem Framing

- List the issue that needs to be resolved.
- Define the project's potential value.
- Determine the project's risks, taking ethical issues into account.
- Create and distribute a flexible, high-level project plan.

# Data Science Process

- Data Acquisition and Preparation

- Selecting data related to the problem.
- Combining the data sets, you may integrate the data.
- Clean the data to find the missing values.
- Handle the missing values by removing or imputing them.
- Errors are dealt with by being removed.
- Use the box plots for detecting outliers and handling them.

## STEP FOR PREPARING THE DATA



# Data Science Process

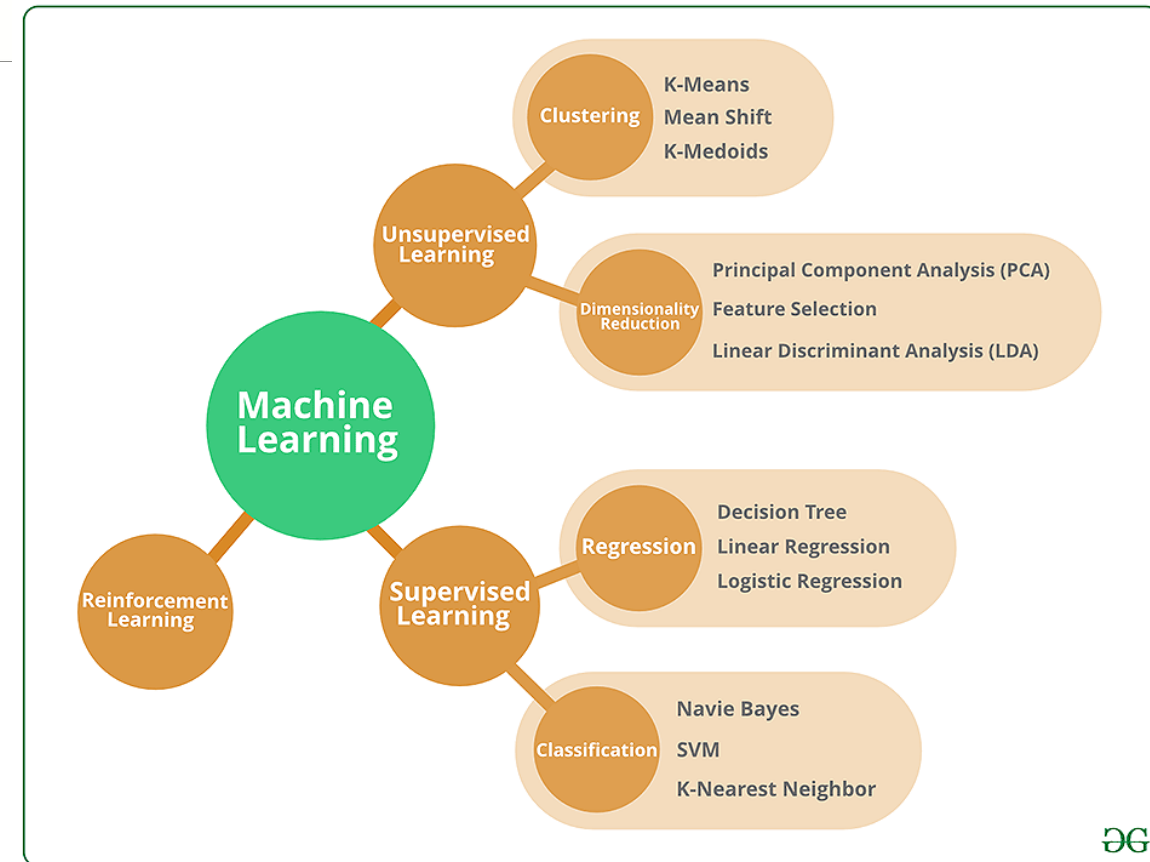
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- Exploratory Data Analysis (EDA)
  - Before creating the real model, it is necessary to get some understanding of the solution and the variables that impact it.
  - To understand data and data features better, we create heat maps, bar graphs, and charting.
  - Additionally, we need to be sure that we recognize the crucial factors in the data set and eliminate any extraneous noise that can really reduce the accuracy of our conclusions.

# Data Science Process

- **Model Planning and Building**

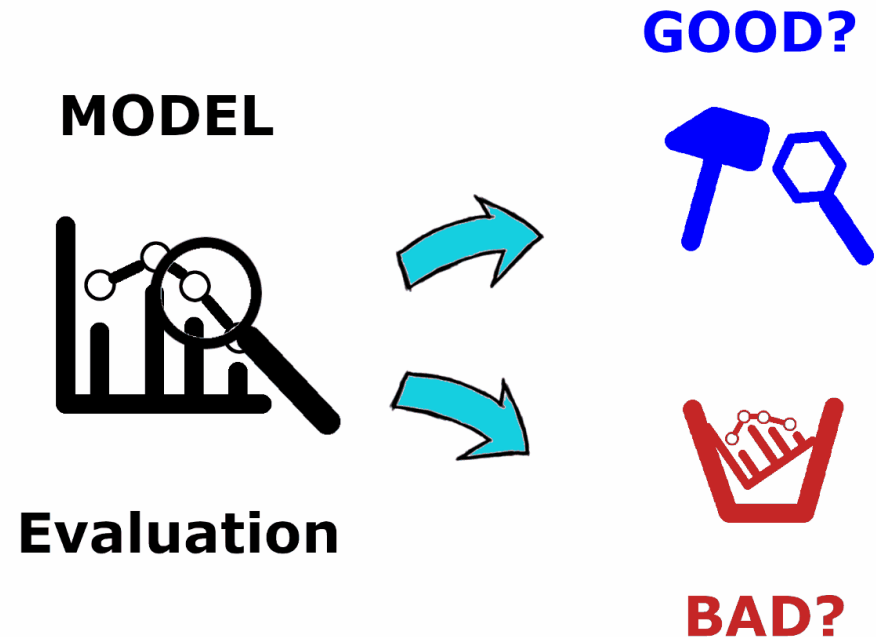
- This phase is about selecting the right model type, depending on whether the issue is classification, regression, or clustering.
- To avoid model overfitting, then hyperparameter tuning is important in model building. This hyperparameter tuning makes the model predict correctly.
- Evaluating the model



# Data Science Process

- Evaluating the model

- Data scientists use either the hold-out or cross-validation technique to assess the efficacy of a model.
- Holdout assessment is used to put a model to the test on data that was not used during training.
- Cross-validation is the process of splitting the data into sets and using them to analyze the performance of the data.



R.Brilenkov

[https://medium.com/@zuhailinasir\\_26370/data-science-process-ad8d1978377e](https://medium.com/@zuhailinasir_26370/data-science-process-ad8d1978377e)

**Question:** What should be done when the evaluation does not yield a satisfying outcome?

# Data Science Process

- **Visualization and Communication**
  - You need to elaborate your findings in a more Humanly manner.
  - And the best way to do that is to use Visualization Graphs and to Communicate the Results in a language, which was used to frame the problem at first.



Example of Data Visualization

<https://datasciencedojo.com/blog/communication-data-visualization/>

# Data Science Process

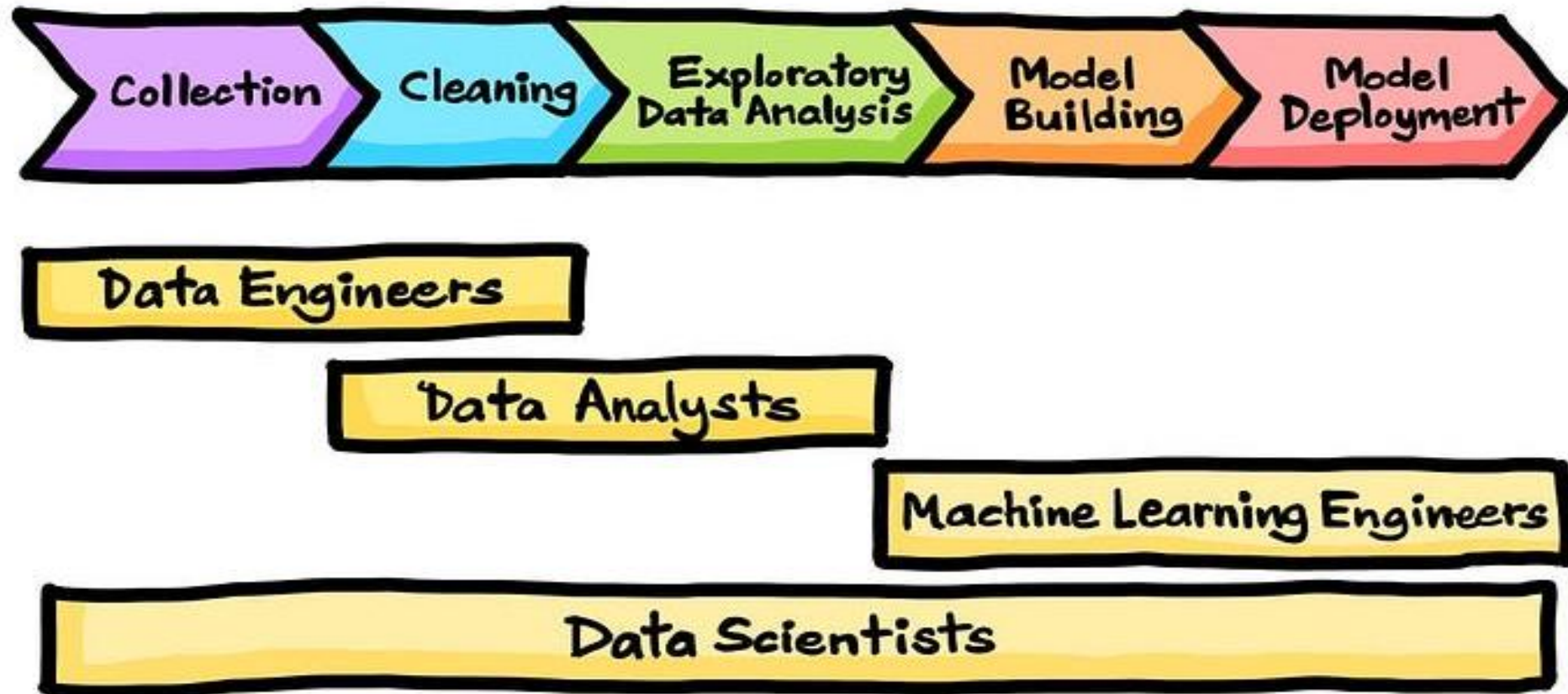
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- Deploying the model

- In this phase, the delivery method that will be used to distribute the model to users or another system is created.
- The process of bringing the model to the cloud and making it accessible.
  - Apply the model with the real data of the company
  - Observe the results from the model
  - Support and maintenance
  - plan to upgrade, update, and fix the model

# Different Individuals Involved in Data Science Projects

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# Different Individuals Involved in Data Science Projects

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- Data Engineers

- The right data are made available and accessible by data engineers for data science projects. They create, develop, and code programs that are data-focused and collect and clean data.

# Different Individuals Involved in Data Science Projects

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- Data Analysts

- A data analyst reviews data to identify key insights into a business's customers and interpret data in order to solve a specific problem. The role includes plenty of time spent with data but entails communicating findings too.

# Different Individuals Involved in Data Science Projects

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- Machine Learning Engineers

- Large data analytics programs are designed, created, and coded by Machine Learning Engineers to assist scientific or business/enterprise activities.
- This position enables models to be deployed (i.e., used in production) and calls for some data science expertise as well as practical software development knowledge.

# Different Individuals Involved in Data Science Projects

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- Data Scientists

- Finding and analyzing rich data sources, combining data sources, developing visualizations, and utilizing machine learning to build models that help derive practical insight from the data are all tasks performed by data scientists.

# Summary

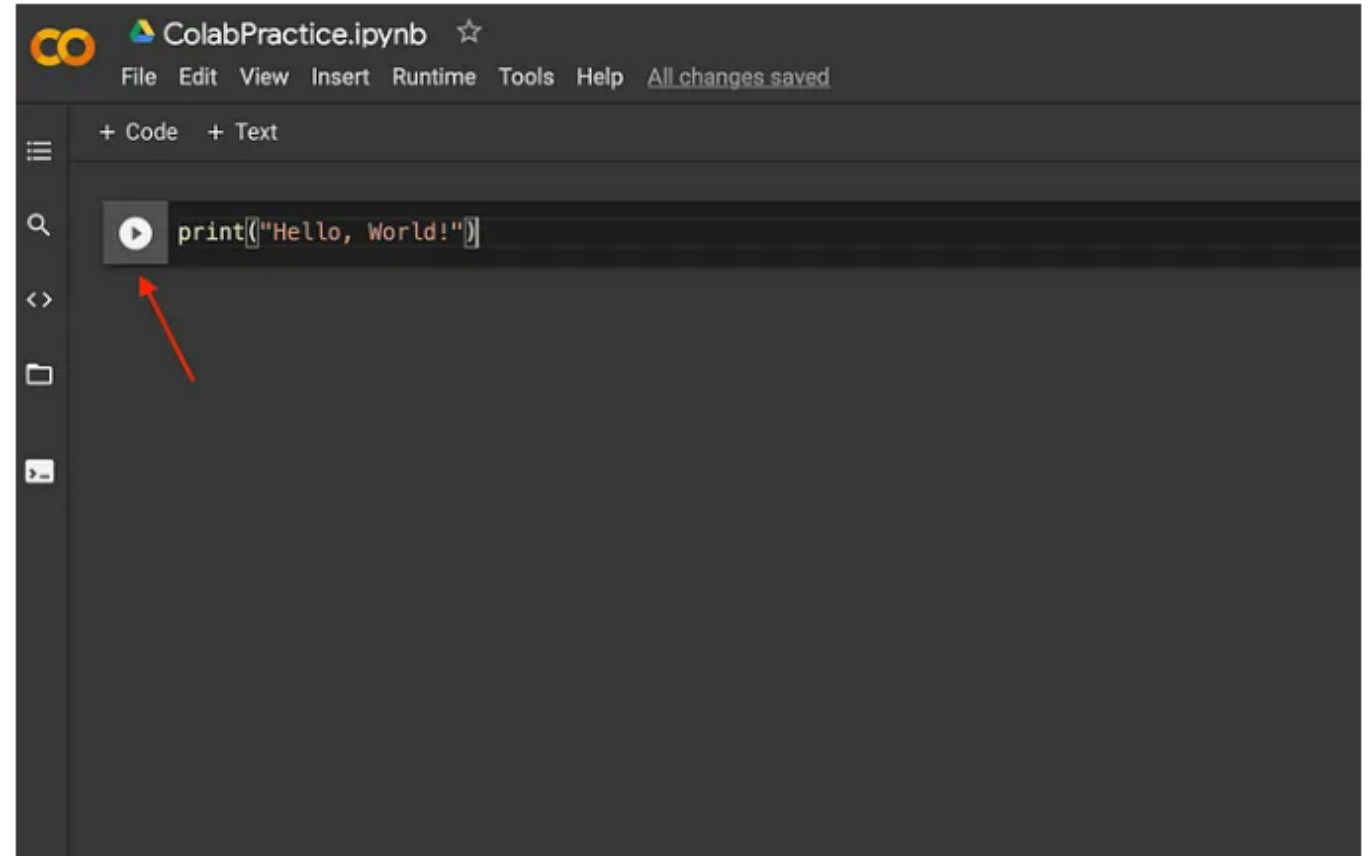
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- Data Scientist
  - The Sexiest Job of the 21<sup>st</sup> Century
- They find stories, extract knowledge. They are not journalists!!!



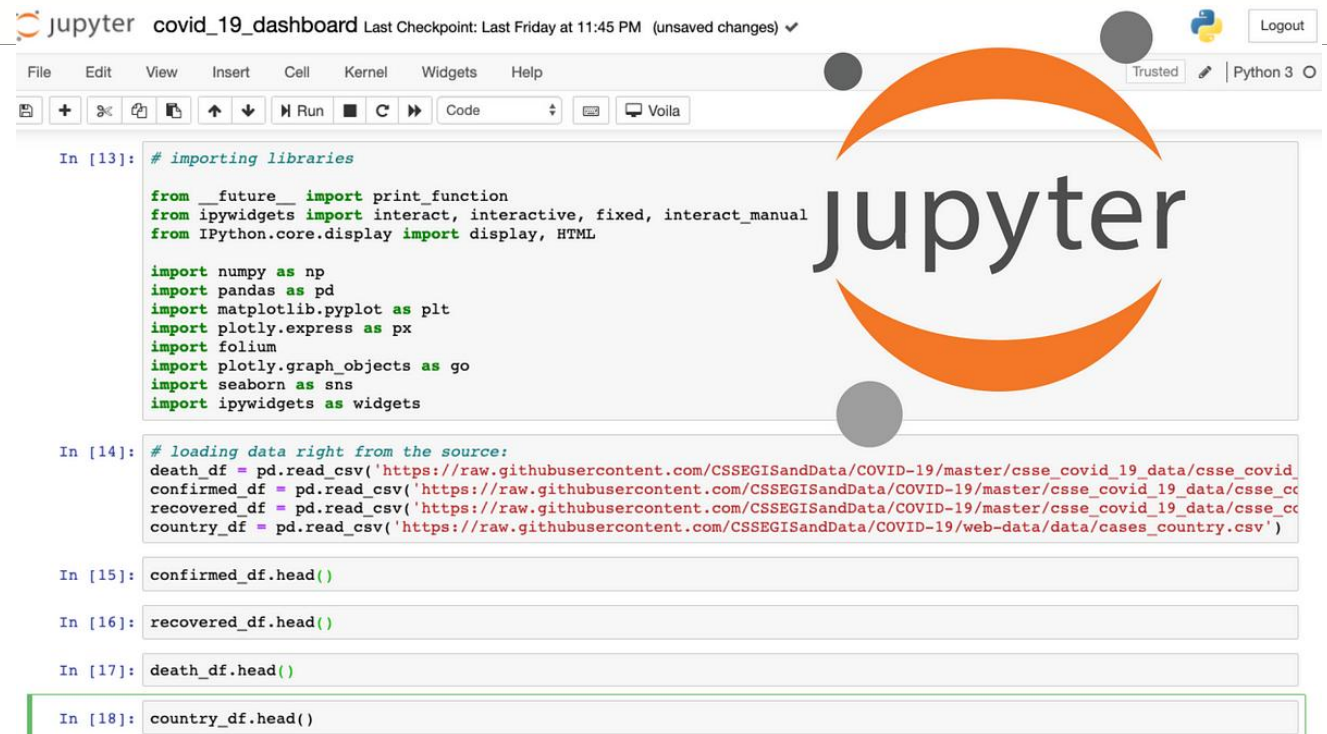
# Laboratory

- Google Colab
  - It is a collaborative software development platform that offers an easy way to develop applications using a variety of coding languages, including python.
  - Google Colab helps users collaborate on code and content management via the Google Cloud Platform (GCP).



# Laboratory

- Jupyter Notebook
  - Jupyter Notebooks are **generally written in python**, which makes them easy to read and edit by non-techies.
  - They can also be **run on an individual's laptop or mobile device**, making them ideal for data scientists working on various machines.



```
In [13]: # importing libraries

from __future__ import print_function
from ipywidgets import interact, interactive, fixed, interact_manual
from IPython.core.display import display, HTML

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px
import folium
import plotly.graph_objects as go
import seaborn as sns
import ipywidgets as widgets

In [14]: # loading data right from the source:
death_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/death_df.csv')
confirmed_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/confirmed_df.csv')
recovered_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covid_19_data/recovered_df.csv')
country_df = pd.read_csv('https://raw.githubusercontent.com/CSSEGISandData/COVID-19/web-data/data/cases_country.csv')

In [15]: confirmed_df.head()

In [16]: recovered_df.head()

In [17]: death_df.head()

In [18]: country_df.head()
```

<https://towardsdatascience.com/the-complete-guide-to-jupyter-notebooks-for-data-science-8ff3591f69a4>

# Practice: Data Science with Python

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- Introduction to Google Colab
  - Functions
  - Loops
  - List Comprehension

